Summary in Graph

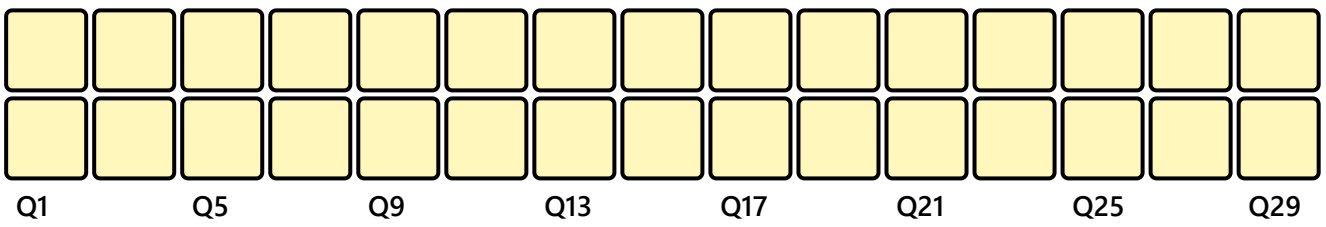
Exam Summary (GATE Overflow CS Test Series 2026 | Operating Systems | Shallow Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (2) out of 22 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Operating System

Which of the following scenarios best illustrates the Convoy Effect? (Mark all the appropriate choices)

A. A high-priority process preempts the CPU from a group of low-priority processes, causing starvation of the lower-priority processes.
B. A set of processes with varying priorities are executed in order of arrival, and the longest process delays all subsequent processes.

C. A process that requires a resource locks it for a long time, causing other processes to wait unnecessarily, even when they could otherwise execute.
D. One slow process slows down the performance of the entire set of processes, and leads to wastage of CPU time and other devices.

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements are true regarding the degree of multi-programming in an operating system?
(Mark all the appropriate choices)

A. The degree of multi-programming increases as more processes are loaded into main memory.
B. A higher degree of multi-programming always results in better CPU utilization.
C. The degree of multi-programming is constrained by the size of physical memory.
D. The degree of multi-programming is determined solely by the short-term scheduler.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements are correct regarding threads?
(Mark all the appropriate choices)

A. Threads within the same process share the same address space and file descriptors.
B. Thread synchronization mechanisms, like mutexes, prevent context switching between threads.
C. Threads provide a mechanism for concurrent execution within a single process.
D. Threads in a process can be executed concurrently but cannot execute in parallel on multiple CPUs unless the operating system supports multi-core scheduling.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements about CPU scheduling algorithms is/are TRUE?
(Mark all the appropriate choices)

A. Longest Job First (LJF) algorithm can result in a high average waiting time and poor system performance.
B. Round-Robin (RR) algorithm has a fixed time slice, which may lead to a higher average waiting time compared to SJF in cases where processes have significantly different burst times.
C. Shortest Job First (SJF) is optimal for minimizing average turnaround time, but it may cause starvation of longer jobs.
D. Shortest Job First (SJF) guarantees the minimum average turnaround time for a set of processes, regardless of their burst times.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements about deadlocks and deadlock handling strategies in an Operating System are TRUE?
(Mark all the appropriate choices)

A. Deadlock detection algorithms rely on detecting cycles in a Resource Allocation Graph (RAG) and are more effective in systems where the number of processes and resources is relatively small.
B. Deadlock avoidance using the Banker's Algorithm can lead to resource underutilization, especially in systems where the maximum resource requirements are difficult to predict or are highly variable.
C. In the wait-for graph, a cycle indicates a deadlock if there is no process that can eventually release the resources required by others in the cycle.
D. Deadlock prevention using the hold-and-wait condition ensures that processes hold all required resources before beginning execution, which increases the risk of resource starvation.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

In a system using the Banker's Algorithm for deadlock avoidance, there are $n = 10$ processes and $m = 5$ resource types. The algorithm performs a safety check after every resource request to ensure the system remains in a safe state. The safety check involves:

1. Iterating through all n processes to determine if a safe sequence exists, simulating resource allocation and release for each process.
2. For each process, iterating through all m resource types to verify that the processes resource needs can be satisfied.

The system processes $k = 100$ resource requests, and no optimizations are applied. What is the total number of operations required to check for safety after k resource requests?

Note: Assume that for each process and resource combination, a constant number of basic operations is required to perform the checks.

Your Answer: | Correct Answer: 50000:50000 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Two semaphores, S_1 and S_2, are initialized to values X_1 and X_2, respectively. The following operations are performed sequentially: 1. 10 $P(S_1)$ operations (wait) are performed on semaphore S_1. 2. 15 $V(S_1)$ operations (signal) are performed on semaphore S_1. 3. 20 $P(S_2)$ operations (wait) are performed on semaphore S_2. 4. 10 $V(S_2)$ operations (signal) are performed on semaphore S_2. After all the operations are completed, the final values of the semaphores are given as:$S_1 = 8$ and $S_2 = 12$ What is the sum of X_1 and X_2?

Your Answer: | Correct Answer: 25:25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Consider a system using a Resource Allocation Graph (RAG) to track the relationships between processes and resources. The following statements relate to the properties of the RAG:
Which of the following statements are correct? (Select all that apply.)

A. A cycle in the RAG involving only request edges (edges from processes to resources) does not necessarily indicate the presence of deadlock.

B. If a process is part of a cycle in the RAG but is not holding any resources, the system cannot reach a potential state of deadlock.
C. The presence of a cycle in the RAG with both request and allocation edges is a necessary condition for the possibility of deadlock, but it is not sufficient to guarantee deadlock.
D. If a RAG contains no request edges, no process can be blocked, and therefore deadlock is impossible.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Consider the following states of processes in an operating system. Match each process state with its corresponding description:

State	Description
$a.$ Suspend Ready	$i.$ The process is in Secondary memory but cannot execute until a condition is satisfied (e.g., resource availability).
$b.$ Suspend Wait	$ii.$ The process is temporarily removed from the ready queue due to resource constraints or other reasons.
$c.$ Ready	$iii.$ The process is waiting in the ready queue, available to execute when a processor becomes available.
$d.$ Blocked	$iv.$ The process is waiting for an event to occur (e.g., I/O completion or a signal).

Which of the following matchings is correct?

A. $a - ii, b - i, c - iii, d - iv$

B. $a - i, b - ii, c - iii, d - iv$

C. $a - ii, b - iv, c - iii, d - i$

D. $a - iii, b - v, c - ii, d - iv$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. A thread in a process has its own stack but shares memory space and file descriptors with other threads in the same process.
B. The operating system kernel is responsible for managing both processes and threads, but the creation of a thread is typically faster than creating a new process.
C. When a thread terminates, it releases all of the resources associated with the entire process.
D. A process can create another process, but a thread cannot create other threads.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. In multi-threaded applications, each thread has its own stack, but all threads within a process share the same heap.
B. The Text section is used to store the program?s dynamic state, such as the current values of variables, which change during execution.
C. When a thread within a process is blocked, it can still execute instructions from other threads within the same process, as long as those threads are not blocked and there are available resources.
D. A process can create multiple threads, but if one of the threads exits, the process will be terminated automatically, regardless of whether the other threads are still running.

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code:

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    if (fork() || fork())
        fork();
    printf("GATE 2025\n");
    return 0;
}
```

The number of times $\mathrm{GATE\ 2025}$ will be printed is _____

Your Answer: | Correct Answer: 10:10 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code :

```
if(fork() == 0) {
    if(fork() == 0) {
        printf("GATE2025");
    }
}
```

The number of times $GATE2025$ will be printed is _____

Your Answer: | Correct Answer: 1:1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following segment of code:

```
int main() {
    pid_t pid1, pid2;
    pthread_t thread;
    pid1 = fork1();
    if (pid1 == 0) {
        fork2();
        pthread_create(...);
        pthread_join(...);
    } else {
        pid2 = fork3();
        if (pid2 == 0) {
            pthread_create(...);
            pthread_join(...);
        }
    }
    fork4();
    return 0;
}
```

If a is the total number of processes created, and b is the total number of threads created when the above code is executed, the value of $b - a =$ _____

Your Answer: | Correct Answer: 3:3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15

Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code:

```
int bar(int n) {
    int i, j;
    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if ((i + j) % 2 == 0) {
                fork();
            }
        }
    }
    return 0;
}
```

The total number of child processes created if bar is called with parameter $n = 4$ is _____.

Your Answer: | Correct Answer: 3:3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following synchronization construct using binary semaphores for two processes:

```
Semaphore mutex = 0;
Process 1
...
while(true)
{
    signal(mutex);
    wait(mutex);
    /* Critical Section */
}
...
Process 2
...
while(true)
    wait(mutex);
    /* Critical Section */
    signal(mutex);
}
...
```

Which of the following statements is/are TRUE?

A. The proposed solution guarantees mutual exclusion but may suffer from deadlock.
B. The proposed solution guarantees mutual exclusion and prevents deadlock.
C. The proposed solution fails to guarantee mutual exclusion and may suffer from deadlock.
D. The proposed solution guarantees mutual exclusion but may not suffer from starvation.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following two-process synchronization solution.

```
P0:
while(TRUE) {
  lock = 1;
  while(lock != 1);
  critical_section();
  lock = 0;
  noncritical_section();
}

P1:
while(TRUE) {
  lock = 1;
  while(lock != 1);
  critical_section();
  lock = 0;
  noncritical_section();
}
```

The shared variable lock is initialized to 0.
Which one of the following is TRUE?

A. This is a correct two-process synchronization solution satisfying mutual-exclusion, progress, and bounded waiting.
B. This solution violates mutual exclusion requirement.
C. This solution violates progress requirement.
D. This solution violates bounded wait requirement.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following synchronization solution.

```
bool flag0 = FALSE;
bool flag1 = FALSE;
P0:
do
{
    flag0 = TRUE;
    while (flag1 == TRUE);
    //Critical Section
    flag0 = FALSE;
} while(TRUE);
P1:
do
{
    flag1 = TRUE;
    while (flag0 == TRUE);
    //Critical Section
    flag1 = FALSE;
} while(TRUE);
```

The two processes $P0$ and $P1$ use $flag0$ and $flag1$ to coordinate access to the critical section. The shared flags $flag0$ and $flag1$ are initially $FALSE$. All assignment statements **(flag0 = TRUE, flag0 = FALSE, etc.)** are atomic in hardware. Which of the following statements is/are correct?

A. Race condition may occur.

B. Mutual exclusion is guaranteed.

C. Progress is guaranteed.

D. Starvation freedom is guaranteed.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following two process synchronization solution. Assume all assignments are atomic.

```
bool flag[2] = {false, false};
int turn;
P0:
do {
    flag[0] = true;
    turn = 0;
    while (flag[1] == true || turn == 1);

    /*Critical section*/
    flag[0] = false;
    // Remainder section
} while (true);
P1:
do {
    flag[1] = true;
    turn = 1;
    while (flag[0] == true || turn == 0);

    /*Critical section*/
    flag[1] = false;
    // Remainder section
} while (true);
```

Which of the following is/are correct? (Mark all the appropriate choices)

A. Deadlock is present.
B. Mutual exclusion is guaranteed.
C. Progress is guaranteed.
D. Bounded waiting is guaranteed.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Fetch_And_Add is an atomic Read-Modify-Write instruction that reads the value of memory location say X, and increments it by the value say i, and returns the old value of X. It is used in the pseudocode shown below to implement a ticket lock. yield() is a method that causes the current thread to give up execution to another thread that is ready to run. A **ticket lock** is a synchronization mechanism that uses "tickets" to control which thread of execution is allowed to enter a critical section. Initially, $turn = ticket = 0$

```
typedef struct {
    int ticket;
    int turn;
} lock_t;

void lock_init(lock_t *lock) {
    lock->ticket = 0;
    lock->turn = 0;
}

void acquire(lock_t *lock) {
    int myturn = Fetch_And_Add(&lock->ticket);
    while (lock->turn != myturn)
          yield();
}

void release(lock_t *lock) {
    lock->turn += 1;
}
```

Following is the implementation of Fetch_And_Add(FAA):

```
    int FAA(int *ptr) {
        int old = *ptr;
        *ptr = old + 1;
        return old;
5. }
```

Suppose multiple threads tries to acquire the ticket lock then for the above pseudocode: (Mark all appropriate choices.)

A. Mutual Exclusion is satisfied

B. Progress is satisfied

C. Bounded Waiting is satisfied

D. Deadlock Exists

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

The following program consists of 4 concurrent processes and 4 binary semaphores. The semaphores are initialized as $S_0 = 0$, $S_1 = 0$, $S_2 = 1$, and $S_3 = 1$.

Process P_0	Process P_1	Process P_2	Process P_3
while (true) {	wait (S_0);	wait (S_2);	wait (S_3);
wait (S_1);	wait (S_1);	wait (S_1);	release (S_1);
print '1';	print '1';	print '1';	print '1';
wait (S_0);	release (S_2);	release (S_3);	release (S_0);
}			

How many times '1' gets printed?

A. At least twice
B. Exactly once
C. Exactly twice
D. None of the above

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following process synchronization program involving N processes:

```
semaphore S = 3;
semaphore T = 2;
Process 1:
S.signal();
T.signal()
while (1)
{
    S.wait();
    T.wait();
    /*critical section*/
    S.signal();
    T.signal();
    //do something
}
Process 2 to Process N:
while (1)
{
    S.wait();
    T.wait();
    /*critical section*/
    S.signal();
    T.signal();
     // do something
}
```

What is the maximum number of processes that can be in the critical section at any moment of time?

A. 1

B. 2

C. 3

D. More than 3

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following code using semaphores:

```
semaphore S1 = 1, S2 = 0, S3 = 1;
int c1 = 1, c2 = 1, c3 = 1;
Process 1
while (c1--) {
  // do something
  S1.wait();
  print{"Start"};
  S3.signal();
  // do something
}
Process 2
while (c2--) {
  // do something
  S3.wait();
  print{"Continue"};
  S2.signal();
  // do something
}
Process 3
while (c3--) {
  // do something
  S2.wait();
  print{"End"};
  S1.signal();
  // do something
}
```

The semaphores are initialized as follows: $S1 = 1$, $S2 = 0$, and $S3 = 1$. The integer variables $c1$, $c2$, and $c3$ are initialized to 1. The output of the above program can be _____.

A. Start Continue End

B. Continue Start End

C. Continue End Start

D. Start End Continue

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following Producer-Consumer implementation using a buffer of size N, where initially the buffer is not empty.

```
Semaphore mutex = __;
Semaphore empty = __;
Semaphore full = __;

Producer {
  while (1) {
    Produce new resource;
    wait(empty);
    wait(mutex);
    Add resource to buffer;
    signal(mutex);
    signal(full);
  }
}
Consumer {
  while (1) {
    wait(full);
    wait(mutex);
    Remove resource from buffer;
    signal(mutex);
    signal(empty);
    Consume resource;
  }
}
```

For the correct functioning of the above code, the initialization of the semaphores $mutex$, $empty$, and $full$ must be:

A. $mutex = 1, empty = N - k, full = k$

B. $mutex = 1, empty = k, full = N - k$

C. $mutex = 0, empty = N - k, full = k$

D. $mutex = 1, empty = N, full = 0$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the classical Dining Philosophers Problem where 7 philosophers are seated at a round dining table. Each philosopher needs 2 chopsticks to eat: one chopstick to the left and one to the right. The chopsticks are shared between adjacent philosophers, and a philosopher must pick up both chopsticks before eating. Semaphores $chopstick[0..6]$ all initialized to 1;
The following pseudocode represents the philosophers behavior:

```
repeat
    wait(chopstick[i]);
    wait(chopstick[(i+1) % 7]);
...
    eat
...
    signal(chopstick[i]);
    signal(chopstick[(i+1) % 7]);
...
    think
...
until false;
```

Which of the following are correct w.r.t. this system? (Mark all the appropriate choices)

A. Deadlock is present
B. Starvation present
C. No Mutual Exclusion

D. Progress is not satisfied.

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a system with k resources of the same type. These resources are shared by 4 processes: P_1, P_2, P_3, and P_4, which have peak demands of 6, 8, 5, and 7 resources respectively. Further, there is a restriction that at any point in time, processes P_1 and P_2 together cannot hold more than 13 resources, while no such restrictions apply to P_3 and P_4.
For which of the following values of k, deadlock may occur? (Mark all the appropriate choices)

A. 20

B. 21

C. 22

D. 23

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a system with n processes, each of which is requesting and holding instances of resources of type R. Let x_i denote the number of resources currently held by process i, and y_i denote the number of resources requested by process i.
At any point in time, the system can be in a safe state if it is possible to allocate resources in such a way that every process can eventually obtain the required resources and complete its execution, without resulting in a deadlock.
Which of the following conditions must hold for the system to be in a safe state?

A. The sum of all currently held resources must be greater than or equal to the sum of all requested resources.
B. For any process i, the total number of resources it is requesting (y_i) must be less than the sum of resources available in the system and the resources held by other processes.
C. The total number of resources requested by all processes is less than or equal to the total number of resources in the system.
D. There exists at least one process that can complete with the available resources, and after completion, its resources can be released to allow other processes to proceed.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a system with 5 processes P_1 through P_5, and 3 resource types: A (8 instances), B (6 instances), and C (10 instances).
Snapshot at time T_0:

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P_1	3	4	3	6	4	3	4	2	3
P_2	3	1	2	7	2	5			
P_3	1	1	1	5	3	4			
P_4	2	0	1	4	2	4			
P_5	0	4	4	3	4	6			

The system is currently in a safe state. Now, consider the following independent requests for additional resources in the current state:
REQ1: P_1 requests 2 units of A, 0 units of B, and 0 unit of C
REQ2: P_2 requests 1 unit of A, 0 units of B, and 2 units of C
REQ3: P_5 requests 2 units of A, 3 units of B, and 0 units of C
Which of the following statements is true?

A. Only REQ1 and REQ2 can be permitted.
B. Only REQ2 and REQ3 can be permitted.
C. All of REQ1, REQ2, and REQ3 can be permitted.
D. None of REQ1, REQ2, or REQ3 can be permitted.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A system has five processes and four allocatable resources. The current allocation and maximum needs are as follows:

	Allocation				Maximum				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P_0	1	0	2	1	3	2	2	2	1	1	x	2
P_1	1	2	1	1	2	3	2	2				
P_2	2	1	1	1	3	3	2	2				
P_3	1	1	2	1	2	2	3	2				
P_4	0	2	1	1	2	3	2	2				

The smallest value of x for which the given state is a safe state is:

A. 1
B. 2
C. 3
D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following set of processes, with their arrival times and burst times given in milliseconds:

Process	Arrival Time	Burst Time
P_1	0	8
P_2	1	4
P_3	3	3
P_4	4	2
P_5	5	5

Using the Preemptive Shortest Remaining Time First (SRTF) scheduling algorithm, calculate the average waiting time for the processes.

Your Answer: | Correct Answer: 4.8:4.8 | Not Attempted | Time taken: 00min 02sec | Discuss

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